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Camshaft and Tappet Reuse Guidelines

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General Information for Cast Iron Camshafts

This bulletin applies specific inspection criteria for midrange engines with chilled ductile iron or chilled cast iron camshaft material with the flat-sliding tappets. The photographs will help as reference tools for inspection and comparison of the sample cam lobes and for ranking the tested cam lobes and for determining the percentage of the useful life left on the camshaft and tappets.

This bulletin addresses the deterioration of cam lobe surfaces at the ramps, edges, and center of the lobes. Cummins Distributors and Dealers are encouraged to use their technical expertise to implement the guidelines for low-cost and high-quality repair of Cummins Inc. products.

The following are definitions of camshaft deterioration:

Pitting:

- Loss of a piece of material from the lobe, resulting in a hole that is visible to the naked eye. Pits typically have rough, dark bottoms and sharp edges and usually occur in the heavily loaded areas of the cam nose or near the edges of the cam lobes.

Chatter:

- The axial lines that run parallel with the axial of the cam lobes. The chatters break down and disturb the oil film thickness. It could cause the metal to metal contact between cam lobe and tappet running surfaces, which increases the cam lobe wear rates and potential for cam lobe and tappet pitting failure.

Distress:

- A localized roughening of the surface that does **not** extend to any significant distance below the lobe surface.

Maco-Spalling:

- Very large pitting. A large local area loss of material from the cam lobe surface.

Table 1, Guidelines for Camshaft Deterioration			
Ranking Scale	Condition of Lobe	Reuse	Percentage of camshaft life remaining
0	New cam lobe, never run	No restriction	100%
1	Normal polish, like new	Unconditional reuse	90%
2	Heavy polish wear, few pitting less than 2 mm in diameter	Unconditional reuse	70%
3	Heavy wear and large pitting areas at cam nose area	Conditional reuse ¹	50%
4	Heavy wear and multiple larger pitting areas at cam nose	Replace	Less than 30%
5	Very heavy cam lobe wear and cam lobe heavy pitting and galling	Replace	10% or less

¹Do **not** reuse the camshaft if the engine is being overhauled or if the camshaft is out of the engine for any reason.

This bulletin is a supplement to the applicable engine family troubleshooting and repair manual.

General Information for Tappets

This bulletin applies specific inspection criteria for midrange engines with sliding tappets when running with chilled ductile iron, chilled cast iron, and steel camshaft materials. The photographs will help as reference tools for inspection and comparison of the sample tappets and for ranking the tested tappets and for determining the percentage of the useful life left on the tappets.

This bulletin addresses the deterioration of sliding tappets at the running surface. Cummins Distributors and Dealers are encouraged to use their technical expertise to implement the guidelines for low-cost and high-quality repair of Cummins products.

The following are definitions of tappet deterioration:

Pitting:

- Same as cam lobe pitting, the running face of the sliding tappet loses one or many small pieces of material from the running face of the tappet, resulting in a hole that is visible to the

naked eye. Pits typically have rough, dark bottoms and sharp edges and usually occur in the heavily loaded areas of the tappet.

Scuffing:

- The running face of the tappet is galling and the material is removing at a high rate. The running surface will lose the material.

Distress:

- A localized roughening of the surface that does **not** extend to any significant distance below the tappet surface.

Maco-Spalling:

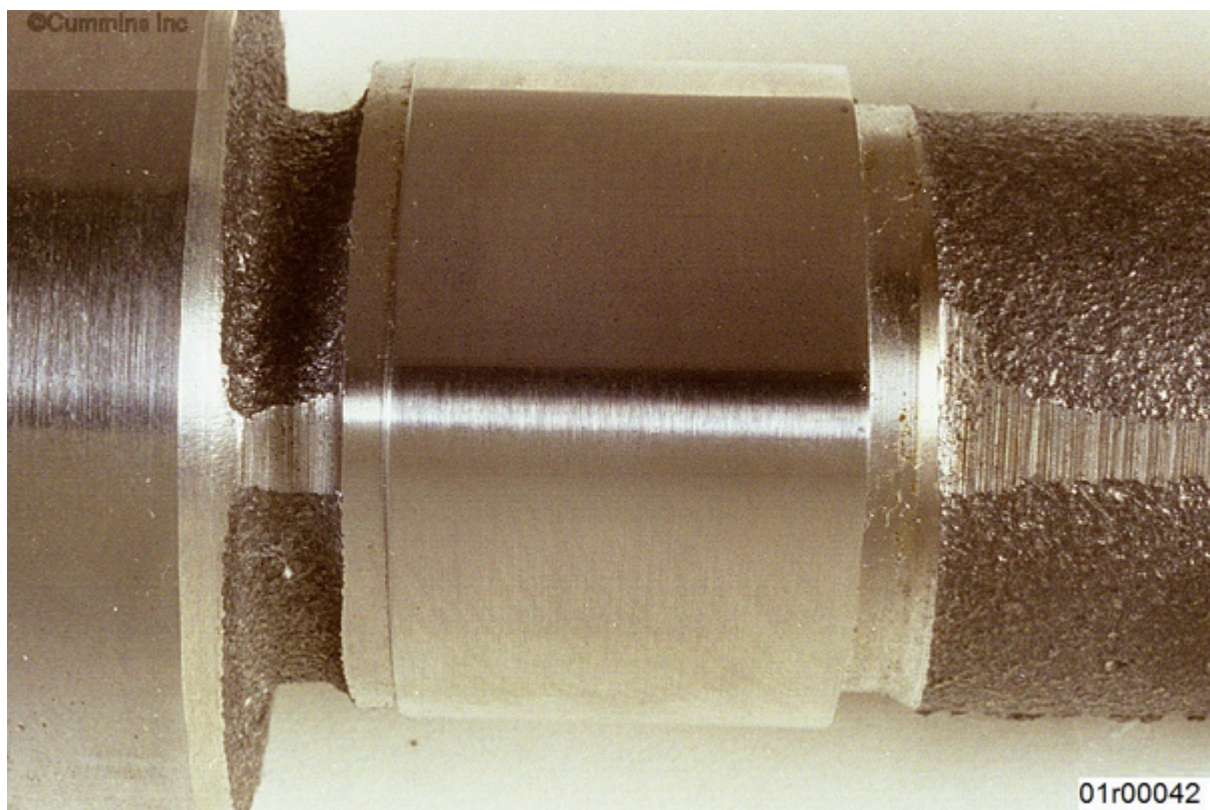
- Very large pitting. A large local area loss of material from the running surface of the sliding tappet.

Table 2, Guidelines for Tappet Deterioration

Ranking Scale	Condition of Tappet	Reuse	Percentage of tappet life remaining
0	New tappet, never run	No restriction	100%
1	Wear polish, like new	Unconditional reuse	90%
2	Normal polish wear, few pitting less than 2 mm in diameter	Unconditional reuse	70%
3	Heavy wear, multiple pitting, wave wear pattern, rough surface distress	Conditional reuse ¹	50%
4	Excessive wear, multiple larger pitting, scuffing, edge chip out, surface porosity.	Replace	Less than 30%
5	Very heavy wear, scuffed, heavy galling, tappet breaking	Replace	10% or less

¹Do **not** reuse the tappets if the engine is being overhauled or if the tappets are out of the engine for any reason.

This bulletin is a supplement to the applicable engine family troubleshooting and repair manual.



Scale 0



Scale 0



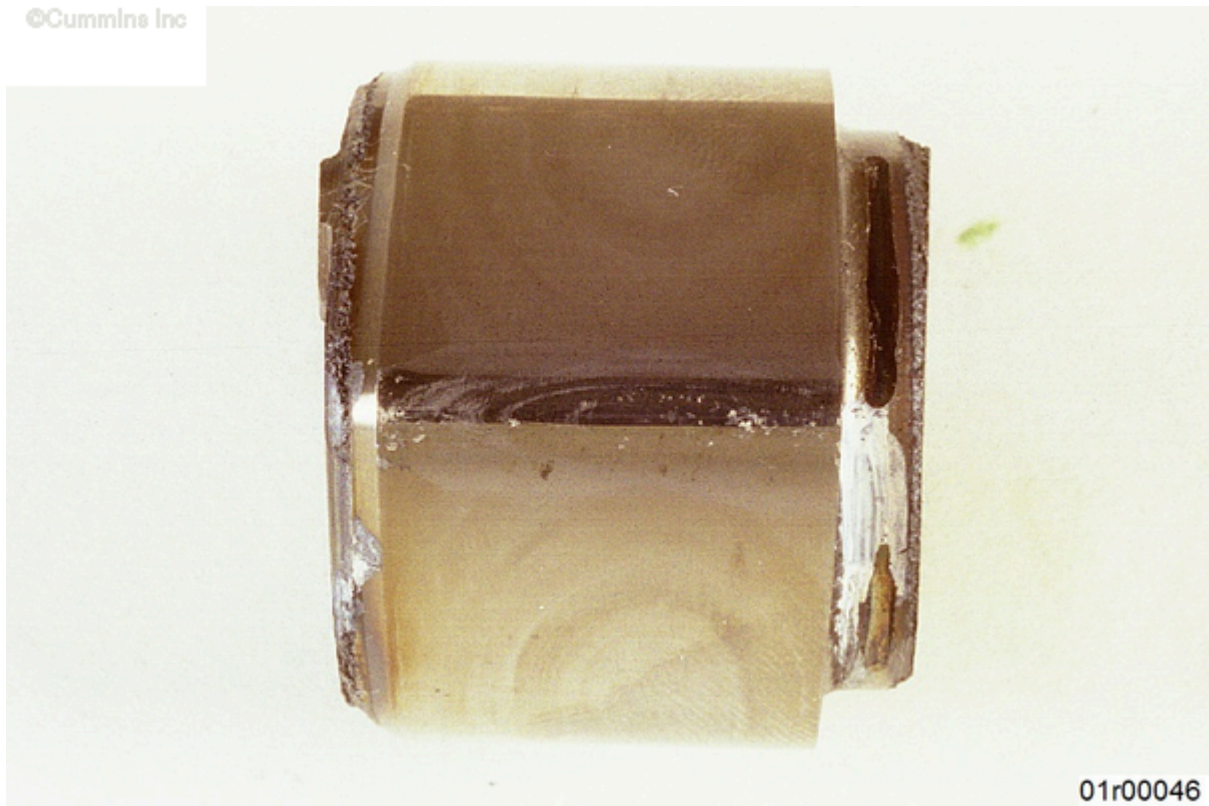
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Scale 1



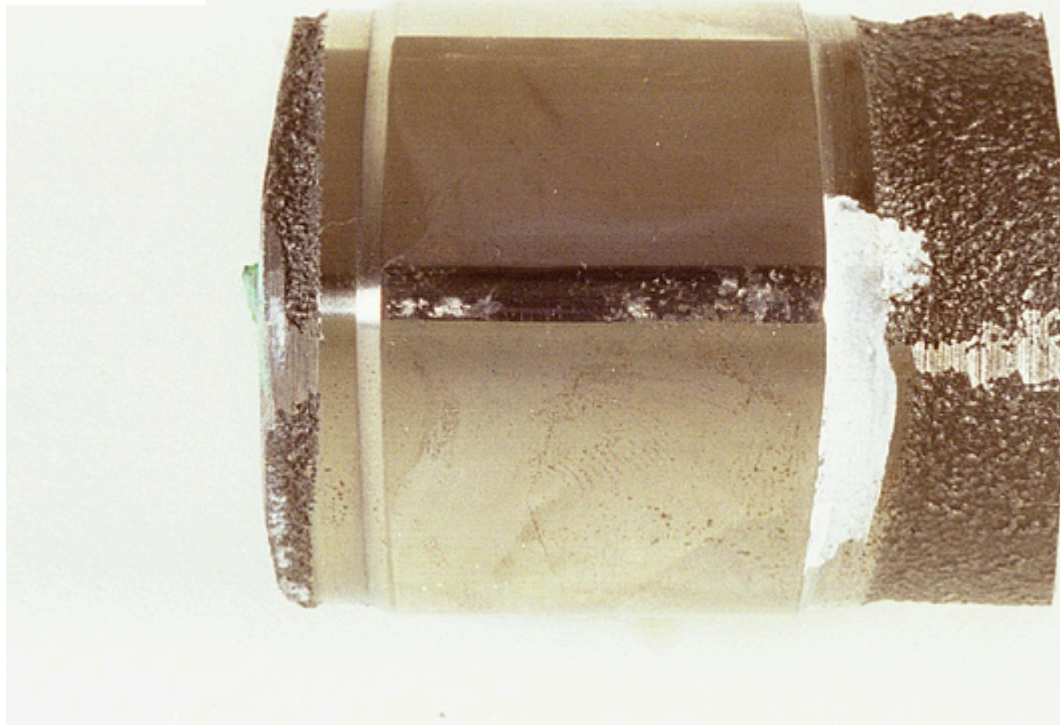
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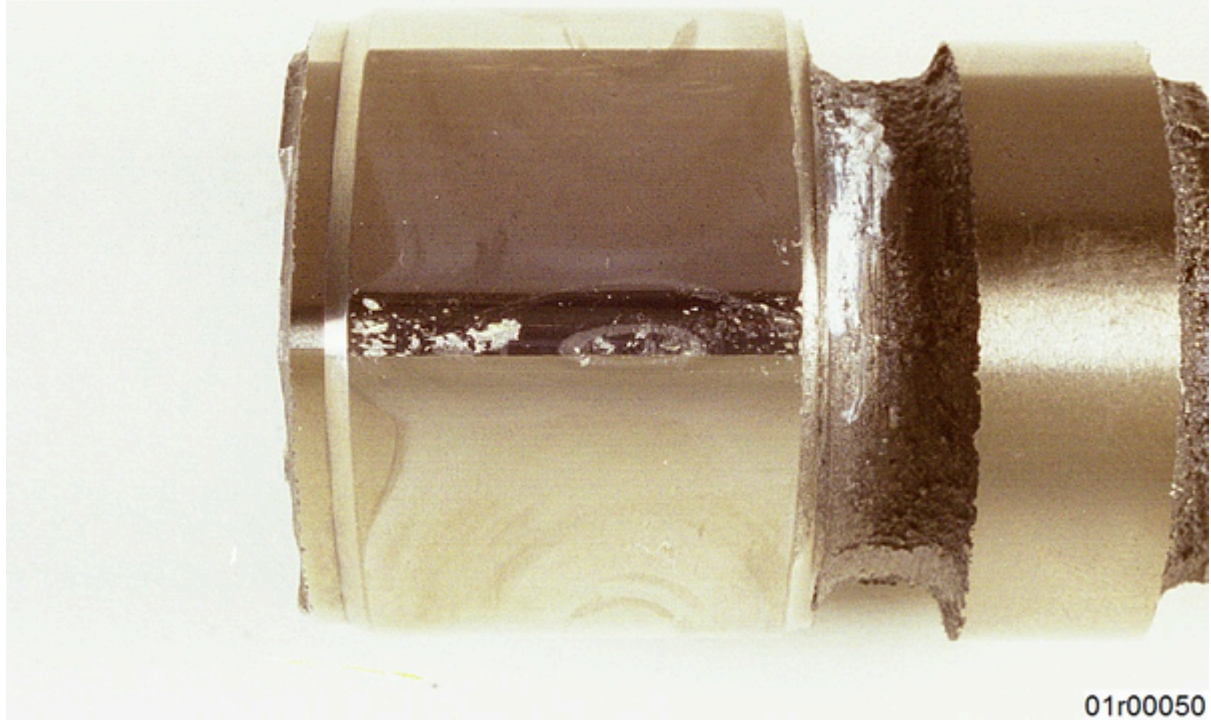
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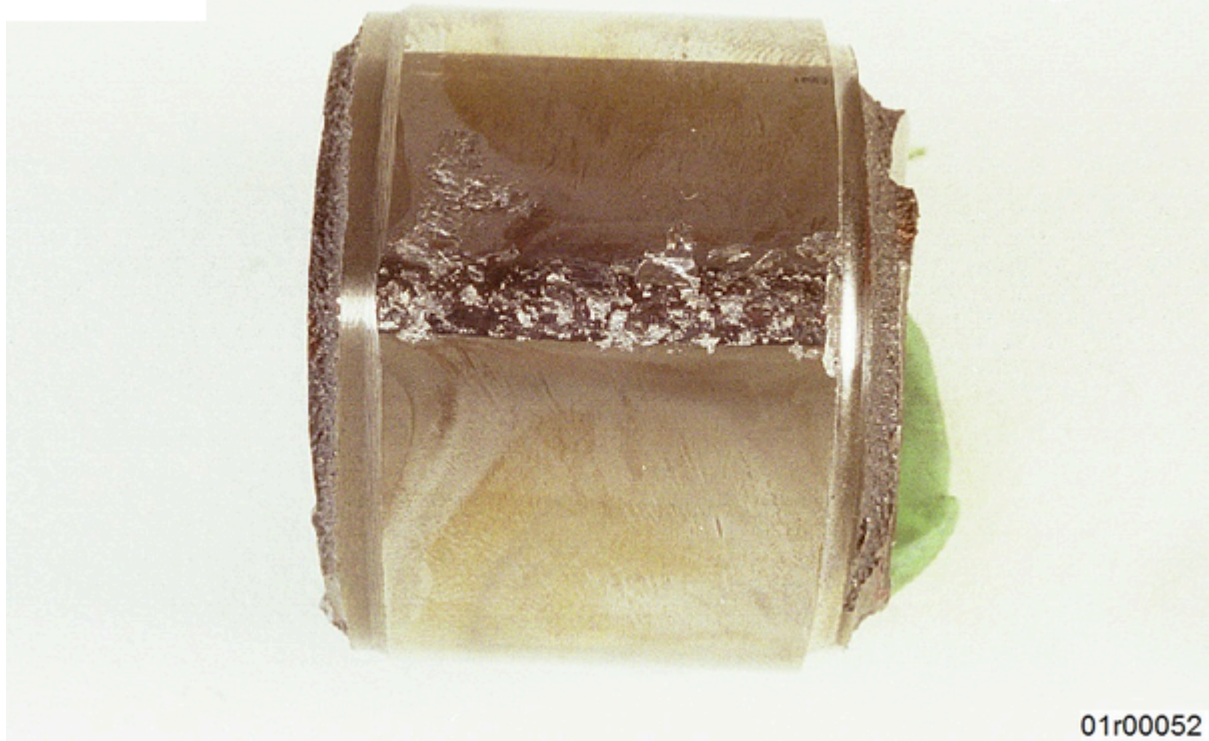
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Scale 5



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